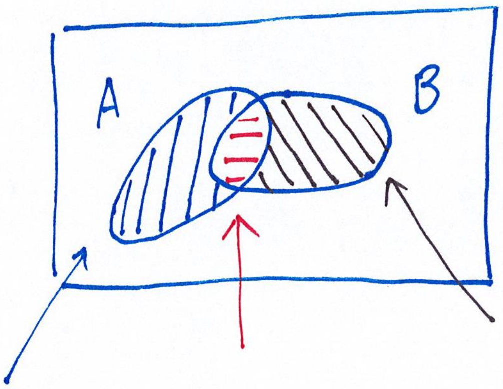
Inclusion and exclusion law

- When A and B are disjoint, then

$$P(A \cup B) = P(A) + P(B)$$

- What if $A \cap B \neq \emptyset$?

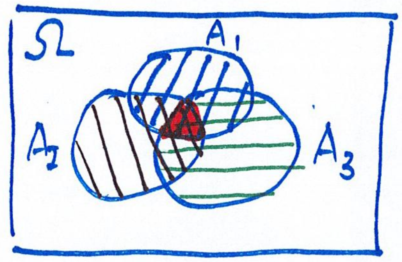


$$\underbrace{(A \setminus B)} \cup \underbrace{(A \cap B)} \cup \underbrace{(B \setminus A)} = A \cup B$$

disjoint

$$\begin{aligned}
\therefore P(A \cup B) &= \boxed{P(A \setminus B) + P(A \cap B)} + P(B \setminus A) \\
&\quad - P(A \cap B) + P(A \cap B) \\
&= P(A) + P(B) - P(A \cap B)
\end{aligned}$$

How about A_1, A_2 and A_3



$$
\begin{aligned}
P(A_1 \cup A_2 \cup A_3) &= P(A_1) + P(A_2) + P(A_3) \\
&\quad - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) \\
&\quad + P(A_1 \cap A_2 \cap A_3)
\end{aligned}
$$

For the most general case:

$$
\begin{aligned}
P\left(\bigcup_{j=1}^{n} A_j\right) &= \sum_{j=1}^{n} P(A_j) - \sum_{i<j} P(A_i \cap A_j) \\
&\quad + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) \\
&\quad \pm \cdots \pm (-1)^{n+1} \cdot P\left(\bigcap_{i=1}^{n} A_i\right)
\end{aligned}
$$

Ex. Roll 3 different 6-sided dice.

Let

$$A_j = \{ j\text{-th dice} = 2 \}, \ j = 1, 2, 3.$$

Find the prob. of having at least 1 dice showing 2.

$$P(\{\text{at least having 1 dice} = 2\})$$

$$= P(A_1 \cup A_2 \cup A_3)$$

$$= P(A_1) + P(A_2) + P(A_3)$$

$$- P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3)$$

$$+ P(A_1 \cap A_2 \cap A_3)$$

We have • $P(A_1) = \frac{1}{6} = P(A_2) = P(A_3)$

• $P(A_1 \cap A_2) = \frac{|A_1 \cap A_2|}{|\Omega|} = \frac{1 \times 1}{6 \times 6} = \frac{1}{36}$

- $P(A_1 \cap A_3) = P(A_2 \cap A_3) = \frac{1}{36}$

- $P(A_1 \cap A_2 \cap A_3) = \frac{|A_1 \cap A_2 \cap A_3|}{|\Omega|}$

$$= \frac{1 \times 1 \times 1}{6 \times 6 \times 6} = \frac{1}{6^3}$$

$\therefore P(A_1 \cup A_2 \cup A_3) = 3 \times \frac{1}{6} - 3 \times \frac{1}{36} + \frac{1}{6^3}$

$$= \frac{91}{216}.$$

Alternatively,

$P(\{\text{having at least 1 dice} = 2\})$

$= 1 - P(\{\text{none of the dice} = 2\})$

$= 1 - \frac{5 \times 5 \times 5}{6 \times 6 \times 6} = \frac{91}{216}.$